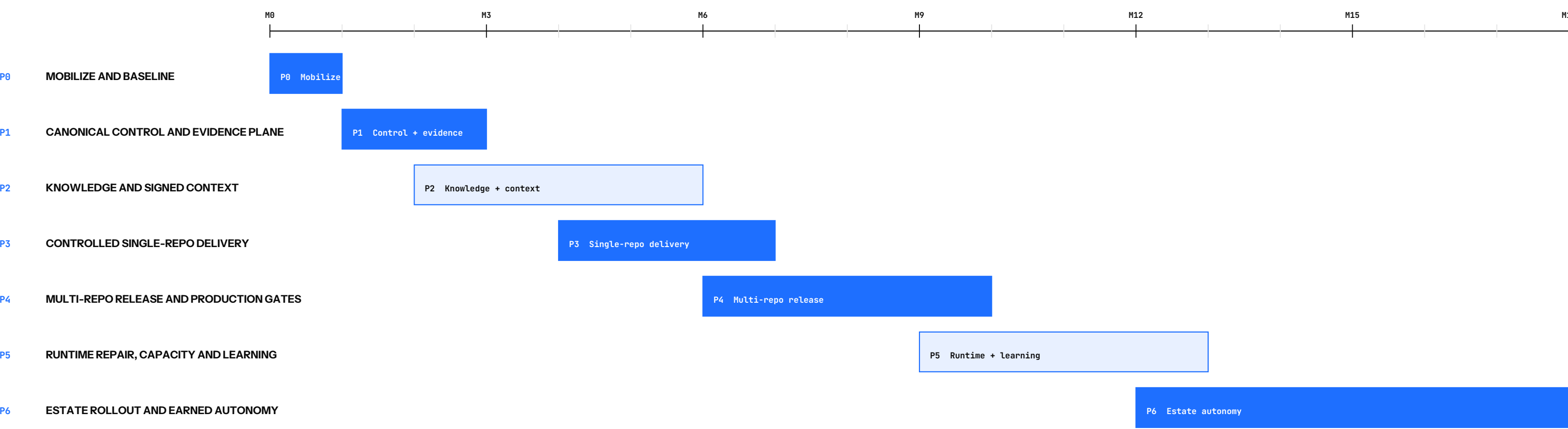


# 18 months to governed autonomy

*Build the control plane first. Earn autonomy by change class. Keep irreversible authority external.*



A / MASTER TIMELINE overlap is intentional; gates remain sequential



B / EXECUTIVE MILESTONES each diamond is a go/no-go decision

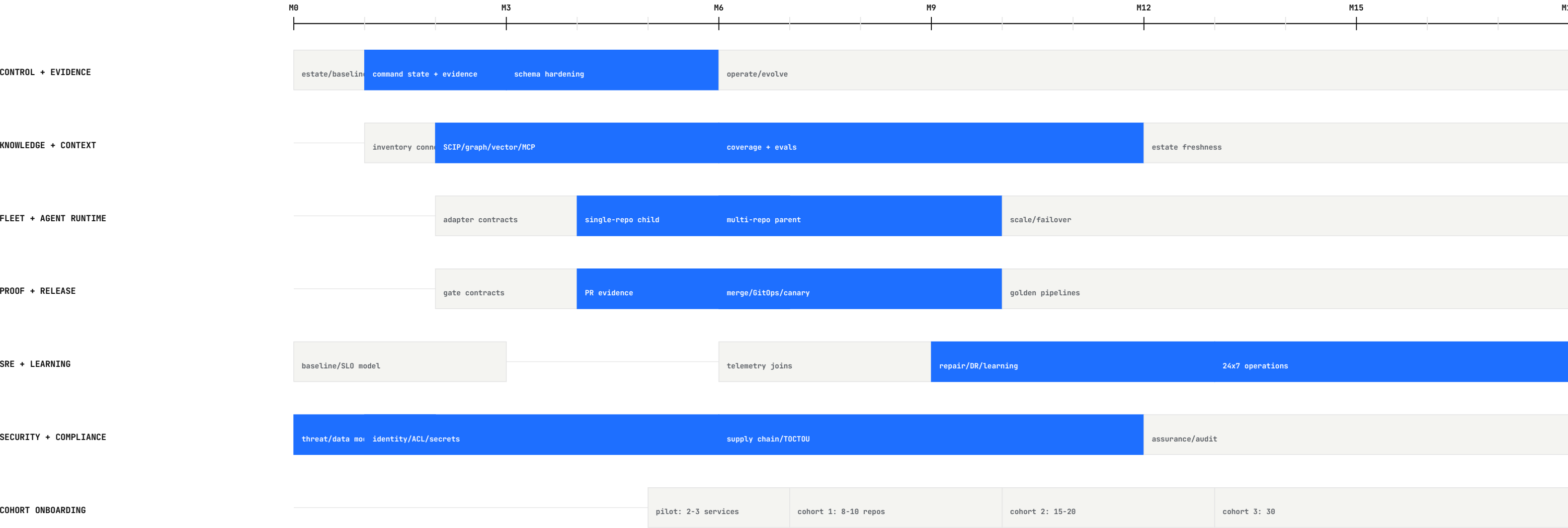


# Critical path and parallel workstreams

*The control/evidence plane gates every other lane; knowledge, runtime and release engineering overlap behind it.*

A / INTEGRATED DELIVERY PLAN

blue bars are critical-path work; gray bars are supporting or cohort work



B / DEPENDENCY RULES

calendar overlap never bypasses an upstream gate

|                                                                              |                                                                                      |                                                                                            |                                                                                              |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>01 NO CONTEXTPACK</b><br>until G1 command/evidence authority is accepted. | <b>02 NO CODE-WRITING CHILD</b><br>until G2 ACL, freshness and retrieval gates pass. | <b>03 NO PRODUCTION CANARY</b><br>until G3 real-task evidence and independent review pass. | <b>04 NO REPAIR AUTONOMY</b><br>until G4 multi-repo compensation and release authority pass. |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|

# P0 / Mobilize and baseline

Month 0 to month 1. One agreed estate, risk model and pilot boundary.

P0



M0 - M1

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

01

**ESTATE MANIFEST**  
30 repos, 100+ services, owners, languages, contracts, data classes, environments, pipelines and SLOs.

02

**BASELINE**  
DORA, change failure, escaped defects, MTTR, CI p95, cloud cost, security backlog and toil.

03

**GOVERNANCE**  
State model, R0-R3 authority, threat model, data retention, pilot cohort and architecture decisions.

PROVE

01

**COVERAGE**  
checked == total for repos, services and production owners; unknowns have named closure dates.

02

**PILOT FITNESS**  
2-3 services include one API, one data change and one real dependency boundary.

03

**DECISION**  
CTO, CISO and platform owners sign scope, no-go rules, staffing and evidence contract.

UNLOCK

01

**CONTROL-PLANE BUILD**  
Connector and evidence work can start against a stable inventory.

02

**PROCUREMENT**  
Sourcegraph, workflow, graph, security and cloud evaluations use measured sizing inputs.

03

**NO AUTONOMY**  
Agents remain research-only; no repository or production writes.

B / EXIT GATE

phase completion is a measured decision, not a status report

G0

Estate accepted: 100% pilot ownership and source coverage, baseline published, risk authority signed.

G0 / HOLD / REDESIGN

# P1 / Canonical control and evidence plane

Month 1 to month 3. Every signal and transition has one authority, one schema and one durable receipt.

P1



M1 - M3

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

- 01

**CONNECTOR SDK**  
Webhook, cursor poll, full snapshot, reconcile and provider writeback for GitHub plus the selected work system.
- 02

**AUTHORITY STACK**  
Postgres command state and outbox, Kafka transport, Temporal workflows, S3 Object Lock evidence.
- 03

**TRUST PLANE**  
OIDC workload identity, OPA transition policy, short leases, tenant binding, audit and break-glass design.

PROVE

- 01

**EVENT TORTURE**  
Duplicate, out-of-order, missing delete, schema drift, replay and connector outage produce no silent loss.
- 02

**STATE SAFETY**  
Illegal transition, stale expected version, expired approval and zero denominator all fail closed.
- 03

**RECOVERY**  
Kill workers and replay events; each side effect occurs once logically and every refusal has a receipt.

UNLOCK

- 01

**CANONICAL CHANGEINTENT**  
Tickets and autonomous findings can enter one versioned lifecycle.
- 02

**EVIDENCE LEDGER**  
Downstream systems can bind decisions to immutable inputs and outputs.
- 03

**STILL NO CODE WRITES**  
The platform may create proposals and ticket comments only.

B / EXIT GATE

phase completion is a measured decision, not a status report

G1

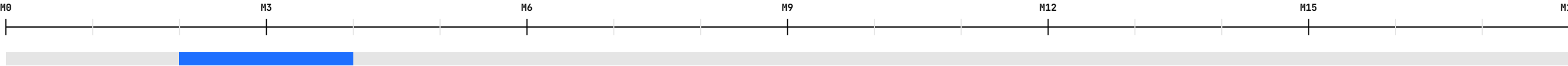
Command path accepted: replayable ingestion, legal transitions, WORM evidence and identity isolation pass adversarial tests.

GO / HOLD / REDESIGN

# P2 / Knowledge and signed context

Month 2 to month 6. Agents receive complete, ACL-safe, reproducible context for an exact change.

P2



M2 - M6

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

01

**CODE INTELLIGENCE**  
Sourcegraph SCIP coverage by repo, commit, root and indexer; AST fallback tagged lower confidence.

02

**KNOWLEDGE PROJECTIONS**  
Neo4j service/code/data/runtime graph; pgvector plus Postgres FTS; tombstones and rebuild pipeline.

03

**CONTEXT GATEWAY**  
MCP read-only retrieval, ACL before candidates, hybrid rank, citations, watermarks, expiry and signature.

PROVE

01

**RETRIEVAL**  
Recall@100 >= 0.95, nDCG@10 >= 0.75, MRR@10 >= 0.85 on protected internal evals.

02

**ISOLATION**  
0 ACL leaks across 10,000 tenant, revoked, poisoned and deleted pairs.

03

**FRESHNESS**  
Commit-to-queryable and permission revocation SLOs publish checked/total; stale context refuses.

UNLOCK

01

**IMPACT PLANNING**  
Planner can trace code, API, event, database, infrastructure and runtime dependencies.

02

**SIGNED CONTEXTPACK**  
Each child receives pinned repo SHAs, graph snapshot, citations, policy and budget.

03

**READ-ONLY AGENT PILOT**  
Agents can draft plans and clarifications using production-shaped context.

B / EXIT GATE

phase completion is a measured decision, not a status report

G2

Context accepted: measured recall, zero ACL leakage, deterministic rebuild and signed freshness proof.

GO / HOLD / REDESIGN

# P3 / Controlled single-repo delivery

Month 4 to month 7. A bounded agent change reaches a protected PR with independent evidence.

P3



M4 - M7

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

- 01

**FLEET CHILD**  
One repo, one worktree, one attempt, pinned image, deny egress, fd3-only worker submission and typed exits.
- 02

**PLAN BEFORE CODE**  
Clarification state, reviewer-first oracle, smallest todos, write-set, exact tests, rollback and budgets.
- 03

**PROOF CONTROLLER**  
Build, unit/integration, Sonar, CodeQL/Snyk/Trivy, secrets, SBOM, provenance and independent review.

PROVE

- 01

**REAL CORPUS**  
20 representative pilot tasks across bug, feature, dependency and failure paths; no stub-only evidence.
- 02

**INDEPENDENCE**  
Implementor cannot edit acceptance tests, self-approve, stamp authority fields or bypass required checks.
- 03

**REPAIR**  
Seed failing gates and bad patches; bounded repair improves evidence or terminates with a typed refusal.

UNLOCK

- 01

**DRAFT PR AUTONOMY**  
R0/R1 agent may open a draft PR; protected systems still own review and merge.
- 02

**PILOT REPOSITORY**  
One low-risk repository can use the full ticket-to-PR path.
- 03

**MEASURED SCORECARD**  
Acceptance, repair, escaped defects, review effort, latency and cost are visible.

B / EXIT GATE

phase completion is a measured decision, not a status report

G3

Single-repo pilot accepted: 20 real tasks, no bypass, independent oracle, bounded repair and exact receipts.

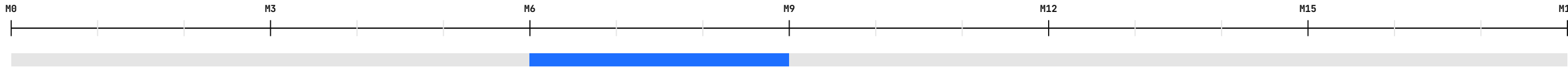
GO / HOLD / REDESIGN



# P4 / Multi-repo release and production gates

Month 6 to month 10. Interdependent changes move as compatibility waves, not a distributed transaction.

P4



M6 - M10

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

- 01

**DEPENDENCY PLANNER**  
Cross-repo DAG, SCC collapse, read/write conflict serialization, contract versioning and release waves.
- 02

**PROTECTED RELEASE**  
Pact can-i-deploy, merge queue, GitOps environment repo, staging, migration gates and canary analysis.
- 03

**ENTERPRISE SECURITY**  
DPoP tool leases, sidecar egress proxy, tenant fences, agent-stack SBOM/SLSA/Cosign and approval epochs.

PROVE

- 01

**SAGA DRILLS**  
10 multi-repo rehearsals include partial merge, rebase, controller loss, migration lock and compensation.
- 02

**RELEASE SAFETY**  
Latest-target checks, signature/provenance validation and rollback canary execute from raw artifacts.
- 03

**AUTHORITY**  
No direct push, stale approval reuse, cross-tenant retrieval or agent-triggered break-glass succeeds.

UNLOCK

- 01

**COHORT 1**  
8-10 repositories onboarded with protected R1 change classes.
- 02

**PRODUCTION PILOT**  
Low-risk compatible changes may canary under deterministic policy.
- 03

**IRREVERSIBLE BOUNDARY**  
Migrations, IAM expansion and destructive actions remain external human authority.

B / EXIT GATE

phase completion is a measured decision, not a status report

G4

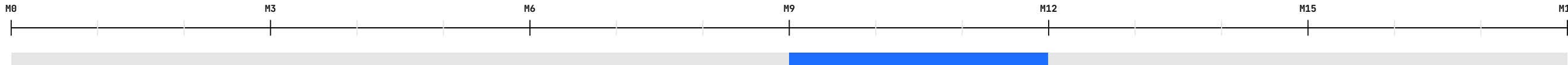
Multi-repo production accepted: 10 saga drills, zero authority bypass, tested compensation and canary rollback.

GO / HOLD / REDESIGN

# P5 / Runtime repair, capacity and learning

Month 9 to month 13. Production evidence re-enters the same governed lifecycle and improves the system.

P5



M9 - M13

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

- 01

**RUNTIME LOOP**  
OTel/CloudWatch RED and USE, SLO burn, deploy correlation, autonomous finding and protected repair PR.
- 02

**CAPACITY AND DR**  
Admission control, WIP/cost quotas, backpressure, regional fences, restore/replay and duplicate-action checks.
- 03

**COMPANY LEARNING**  
Incident evidence becomes a policy, eval, runbook or graph PR with owner, scope, expiry and rollback.

PROVE

- 01

**GAME DAYS**  
10 incident drills cover telemetry loss, noisy alerts, repeated repair, regional loss and stale topology.
- 02

**SLO PROOF**  
Multi-window burn alerts publish numerator/denominator; no-data cannot promote or silently recover.
- 03

**LEARNING QUALITY**  
Seed harmful and stale lessons; poisoning, conflict, expiry and rollback controls reject them.

UNLOCK

- 01

**AUTONOMOUS R1 REPAIR**  
Bounded restart, scale, rollback or code-repair PR may run under signed policy.
- 02

**COHORT 2**  
15-20 repositories and their production telemetry join the lifecycle.
- 03

**OPERATIONAL OWNERSHIP**  
Platform SLOs, pager rotation, runbooks and cost envelopes become mandatory.

B / EXIT GATE

phase completion is a measured decision, not a status report

G5

Operate loop accepted: game days, SLO denominators, DR reconciliation and verified learning promotion pass.

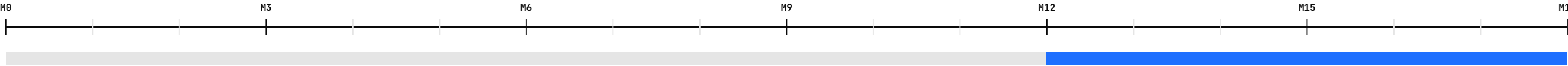
GO / HOLD / REDESIGN



# P6 / Estate rollout and earned autonomy

Month 12 to month 18. The platform is operated as company infrastructure; autonomy is earned per change class.

P6



M12 - M18

A / PHASE CONTRACT

build, falsify, then unlock

BUILD

- 01

**COHORT ROLLOUT**  
Onboarding factory for remaining repositories: contracts, owners, SLOs, policies, evals and golden pipelines.
- 02

**CONTROL OPERATIONS**  
24x7 platform SLO, quota and cost control, model/provider failover, schema evolution and evidence retention.
- 03

**AUTONOMY REGISTRY**  
Per-class scorecard for acceptance, repair, escaped defects, rollback, security, follow-on change and cost.

PROVE

- 01

**OUTCOME WINDOWS**  
At least 60 closed production outcome windows with UNKNOWN and rollback rates explicitly published.
- 02

**ESTATE COVERAGE**  
30/30 repositories mapped; all autonomous classes have owners, policies, evals and kill switches.
- 03

**SUSTAINED CONTROLS**  
No cross-tenant leak, unreceipted refusal, direct merge, stale approval or unknown critical gate.

UNLOCK

- 01

**R1 AT SCALE**  
Reversible low-risk classes may run ticket-to-canary autonomously within quotas.
- 02

**R2/R3 REMAIN PROTECTED**  
Production migration, IAM, destructive data, region failover and legal exceptions stay human-bound.
- 03

**QUARTERLY RE-EARN**  
Autonomy is reduced automatically when scorecards regress; calendar completion grants nothing.

B / EXIT GATE

phase completion is a measured decision, not a status report

G6

Scale accepted: 30 repos, 60 outcome windows, 24x7 ownership and class-level autonomy scorecards remain green.

GO / HOLD / REDESIGN

# Team, decisions and operating constraints

*A platform program, not a tooling rollout. Staff the critical path and force a decision at each gate.*

A / CORE TEAM RAMP

loaded people cost is the dominant controllable investment

M0-M3

8-10 FTE

architecture, control/data, security, platform/SRE, TPM

M3-M9

12-16 FTE

add knowledge/retrieval, Fleet runtime, CI/release and QA/evals

M9-M18

16-22 FTE

add onboarding, 24x7 reliability, enablement and domain integration

B / TEAM TOPOLOGY

one accountable platform owner; federated service owners supply domain truth

CONTROL PLANE

Postgres/Kafka/Temporal/S3, connectors, schemas, evidence

KNOWLEDGE

SCIP, graph, retrieval, Context Gateway, evals

AGENT RUNTIME

Fleet, adapters, isolation, budgets, orchestration

PROOF + RELEASE

CI gates, PR authority, GitOps, canary, supply chain

SRE + SECURITY

SLO/capacity/DR, identity, secrets, assurance, incident learning

DOMAIN COHORTS

0.1-0.2 FTE per participating service team during onboarding

C / EXECUTIVE DECISION CALENDAR

a hold is cheaper than scaling an unproven control

| M1               | M3                | M6                | M7             | M10                   | M13            | M18             |
|------------------|-------------------|-------------------|----------------|-----------------------|----------------|-----------------|
| G0               | G1                | G2                | G3             | G4                    | G5             | G6              |
| scope + baseline | control authority | context integrity | single-repo PR | multi-repo production | operate/repair | estate autonomy |

N0-G0

Any cross-tenant leak, direct merge, unreceipted refusal, unknown critical gate, stale approval reuse, destructive agent action or unverifiable denominator stops autonomy expansion and returns the affected class to proposal-only.